PRESCRIBED GOLDBACH LENGTHS IN THE NONNEGATIVE POLYNOMIAL SEMIRING

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ABSTRACT. We determine the possible numbers of multiplicative atoms in additive decompositions of nonconstant polynomials $f \in \mathbb{N}_0[x]$ with $f(0) \geq 2$. Writing $f = a + bx + P(x)$ with $P \in x^2\mathbb{N}_0[x]$, we prove that f is a sum of exactly $k \geq 2$ atoms if and only if $k \leq a + b$ and $2k \leq a + 2b + P(1)$. The proof is constructive and uses elementary parity criteria. In particular, every such polynomial is a sum of at most two atoms; the only exceptions to a sum of exactly two are $2 + x^r$ with $r \geq 2$. We also record a prescribed-length corollary of Kaplan–Polo's Laurent two-atom theorem. Atomicity is always in the indicated semiring, not in $\mathbb{Q}[x]$.

1. INTRODUCTION

Goldbach-type questions ask for representations as sums of multiplicatively irreducible elements. Hayes [2] and Pollack [6] studied such questions over polynomial rings. In the nonnegative setting, Liao and Polo [5] proved that every $f \in \mathbb{N}_0[x^{\pm 1}]$ with $f(1) > 3$ and $|\operatorname{supp}(f)| > 1$ is a sum of two atoms. Kaplan and Polo [3] recovered this result without prime-distribution estimates and extended it to Laurent polynomial and series semidomains. Related work treats group semidomains [4] and the arithmetic of polynomial semidomains [1].

In the ordinary semiring $\mathbb{N}_0[x]$, the only zero-constant atom is x, whereas x is a unit in the Laurent semiring. Thus a zero-constant polynomial having a term of degree at least two is not a sum of ordinary polynomial atoms. We instead classify the possible numbers of summands when the constant coefficient is at least two.

Throughout, $\mathbb{N} = \{1, 2, \ldots\}$ and $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$. An *atom* is a nonzero nonunit h such that every factorization $h = uv$ in the indicated semiring has a unit factor. The only unit of $\mathbb{N}_0[x]$ is 1. Write $\mathcal{A}$ for its atoms and define

$$\mathsf{L}_+(f) = \{k \in \mathbb{N} : f = h_1 + \cdots + h_k \text{ for some } h_1, \ldots, h_k \in \mathcal{A}\}.$$

Repeated summands and prime-constant atoms are permitted. These are *additive* representation lengths, not multiplicative factorization lengths. Semiring atomicity is also distinct from rational irreducibility: $1 + x^3$ is a semiring atom, although $1 + x^3 = (1 + x)(1 - x + x^2)$ in $\mathbb{Q}[x]$.

Our main result depends only on the constant coefficient, linear coefficient, and total mass of the higher-degree coefficients.

Theorem 1.1 (Prescribed-length classification)**.** *Let $f(x) = a + bx + P(x) \in \mathbb{N}_0[x]$ be nonconstant, where $a \geq 2$, $b \geq 0$, and $P \in x^2\mathbb{N}_0[x]$. Put $m = P(1)$. For every integer $k \geq 2$,*

$$k \in \mathsf{L}_+(f) \iff k \leq a + b \quad \text{and} \quad 2k \leq a + 2b + m. \tag{1}$$

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Equivalently,

$$K(f) = \min\left\{a+b, \left\lfloor \frac{a+2b+m}{2} \right\rfloor\right\}, \tag{2}$$
$$\mathsf{L}_+(f) \cap \{2,3,\dots\} = \{2,3,\dots,K(f)\},$$

where the right side is empty if $K(f) < 2$. Length one occurs precisely when f itself is an atom.

The two inequalities are elementary necessary conditions. Their sufficiency holds for arbitrary degree and coefficient height, with no restriction on the positions of the exponents in P. The ordinary classification has a self-contained proof; the Laurent comparison in Section 5 uses the cited two-atom theorem.

Corollary 1.2 (Sharp two-atom statement)**.** *Every nonconstant $f \in \mathbb{N}_0[x]$ with $f(0) \geq 2$ is a sum of at most two atoms. It is a sum of exactly two atoms if and only if*

$$f \neq 2 + x^r \qquad \text{for every } r \geq 2.$$

The exceptional polynomials $2 + x^r$ are themselves atoms.

2. Elementary atom certificates

For $h = \sum_{j\geq 0} h_j x^j \in \mathbb{N}_0[x]$, define

$$E(h) = \sum_{j \text{ even}} h_j, \qquad O(h) = \sum_{j \text{ odd}} h_j.$$

The constant coefficient contributes to $E(h)$.

Lemma 2.1 (Parity certificates)**.** *Suppose $h \in \mathbb{N}_0[x]$ has positive constant coefficient. Then h is an atom if either*

$$O(h) = 1, \qquad \text{or} \qquad E(h) = 1 \text{ and } h \neq 1.$$

Proof. If $h = uv$ in $\mathbb{N}_0[x]$, then $u(0), v(0) > 0$, and

$$E(h) = E(u)E(v) + O(u)O(v),$$
$$O(h) = E(u)O(v) + O(u)E(v).$$

Both $E(u)$ and $E(v)$ are positive integers. If $O(h) = 1$, exactly one term in the second identity is one. After interchanging u, v, we have $O(u) = 0$ and $E(u) = 1$. Since $u(0) > 0$, this forces $u = 1$. If $E(h) = 1$, the first identity forces $E(u) = E(v) = 1$ and $O(u)O(v) = 0$. A factor with no odd terms has exactly one even coefficient unit, necessarily at degree zero, and is therefore 1. In either case h is a nonunit, so it is an atom. □

Lemma 2.2 (Exponent rescaling)**.** *If $H \in \mathbb{N}_0[x]$ has positive constant coefficient and $D \geq 1$, then $H(x^D)$ is an atom if and only if $H(x)$ is an atom. In particular, $c + x^r$ is an atom for all integers $c, r \geq 1$.*

Proof. Suppose $H(x^D) = uv$ in $\mathbb{N}_0[x]$. Both factors have positive constant coefficient. Every exponent in either factor's support also occurs in the product: multiply that term by the other factor's nonzero constant term and use nonnegativity. Hence every factor exponent is divisible by D, so $u = U(x^D)$ and $v = V(x^D)$ for some $U, V \in \mathbb{N}_0[x]$. This factorization descends to $H = UV$, and units are preserved by substitution. The converse follows by substituting in a factorization of H. Finally, $c + x$ satisfies Lemma 2.1; rescaling gives $c + x^r$. □

Lemma 2.3 (Two-atom construction)**.** *Let $A \geq 2$, let $0 \neq R \in x\mathbb{N}_0[x]$, and suppose $A + R(1) \geq 4$. Then $A + R$ is a sum of two atoms of $\mathbb{N}_0[x]$.*

Proof. If $R(1) = 1$, then $R = x^r$ and $A \geq 3$. The decomposition

$$A + x^r = 2 + \big((A-2) + x^r\big)$$

uses the prime constant 2 and an atom from Lemma 2.2.

Suppose $R(1) \geq 2$. Let D be the largest power of two dividing every exponent in $\operatorname{supp}(R)$. Write $R(x) = T(x^D)$. At least one exponent of T is odd; otherwise $2D$ would divide every exponent of R. Decompose

$$T = T_{\mathrm{e}} + T_{\mathrm{o}},$$

where T_{e} has positive even exponents and T_{o} has odd exponents. Thus $T_{\mathrm{o}}(1) \geq 1$.

If $T_{\mathrm{o}}(1) \geq 2$, choose an odd exponent r occurring in T_{o}. Then

$$A + T = \big((A-1) + x^r + T_{\mathrm{e}}\big) + \big(1 + T_{\mathrm{o}} - x^r\big). \tag{3}$$

The first summand has odd coefficient sum one. The second has even coefficient sum one and is not the unit, because $T_{\mathrm{o}}(1) \geq 2$. Both have positive constant coefficient and are atoms by Lemma 2.1.

If $T_{\mathrm{o}}(1) = 1$, write $T_{\mathrm{o}} = x^r$. Then $T_{\mathrm{e}} \neq 0$, since $T(1) \geq 2$. Choose an even exponent s occurring in T_{e}. We have

$$A + T = \big((A-1) + x^s\big) + \big(1 + x^r + T_{\mathrm{e}} - x^s\big). \tag{4}$$

The first summand is an atom by Lemma 2.2; the second has odd coefficient sum one and is an atom by Lemma 2.1. Every coefficient in these decompositions is nonnegative. Substituting x^D preserves atomicity by Lemma 2.2 and gives the required decomposition of $A + R$. □

3. Proof of the prescribed-length classification

For $h = \sum_{j \geq 0} h_j x^j$, introduce the additive weights

$$U(h) = h_0 + h_1, \qquad W(h) = h_0 + 2h_1 + \sum_{j \geq 2} h_j = h(1) + h_1. \tag{5}$$

Lemma 3.1 (Necessary capacity inequalities)**.** *Every atom $h \in \mathbb{N}_0[x]$ satisfies $U(h) \geq 1$ and $W(h) \geq 2$. Consequently, every sum of k atoms satisfies $U(f) \geq k$ and $W(f) \geq 2k$.*

Proof. The only zero-constant atom is x: a nonzero polynomial with zero constant coefficient factors as xq, and atomicity forces $q = 1$. The constant atoms are exactly the positive integer primes. Any other atom has positive constant coefficient and at least one positive-degree term. For x the weights are $(1, 2)$; for a prime constant they are both at least two; and for a positive-constant nonconstant atom the inequalities are immediate. Additivity proves the assertion for a sum. □

Proof of Theorem 1.1. Necessity follows from Lemma 3.1, since $U(f) = a+b$ and $W(f) = a+2b+m$.

For sufficiency, fix an integer $k \geq 2$ satisfying the two inequalities, and induct on k. If $k = 2$ and $f(1) \geq 4$, apply Lemma 2.3. Otherwise $f(1) = 3$, so $f = 2 + x^r$. The condition $W(f) \geq 4$ forces $r = 1$, and $2 + x$ is the sum of the two atoms 2 and x.

Now let $k \geq 3$. We remove one atom so that the remainder is nonconstant, has constant coefficient at least two, and satisfies the capacity inequalities for $k - 1$.

Case 1: $a \geq k + 1$. Remove 2. The remainder has constant coefficient $a - 2 \geq k - 1 \geq 2$ and stays nonconstant. Moreover,

$$U(f-2) = a + b - 2 \geq k - 1, \qquad W(f-2) = W(f) - 2 \geq 2(k-1).$$

Case 2: $a \leq k$ and $b \geq 1$. Remove x. The constant coefficient is unchanged, and

$$U(f-x) = U(f) - 1 \geq k - 1, \qquad W(f-x) = W(f) - 2 \geq 2(k-1).$$

If $f - x$ were constant, then $f = a + x$, giving $2k \leq a + 2 \leq k + 2$, impossible for $k \geq 3$. Thus the remainder is nonconstant.

Case 3: $a \le k$ and $b = 0$. The first inequality gives $a = k$, and the second gives $m \ge k$. Choose $r \ge 2$ with $[x^r]P > 0$, and remove the atom $1 + x^r$. The remainder has constant coefficient $k - 1 \ge 2$ and higher-degree mass $m - 1 \ge k - 1 > 0$. Its weights satisfy

$$U(f - (1 + x^r)) = k - 1, \qquad W(f - (1 + x^r)) = W(f) - 2 \ge 2(k - 1).$$

In all three cases, the induction hypothesis writes the remainder as a sum of $k - 1$ atoms. Adjoining the removed atom proves sufficiency. Formula (2) follows directly from (1), and the assertion about length one follows from the definition. □

The proof is constructive: repeatedly remove 2, x, or $1 + x^r$ according to the three cases, and finish with the explicit two-atom decomposition. No search for integer prime pairs is required.

4. Sharp consequences in the ordinary semiring

Proof of Corollary 1.2. For $k = 2$, the bound $2 \le a + b$ is automatic. For a nonconstant target with $a \ge 2$, the other inequality fails exactly when $a = 2$, $b = 0$, and $m = 1$, that is, when $f = 2 + x^r$ with $r \ge 2$. These exceptional polynomials are atoms by Lemma 2.2. □

Corollary 4.1 (Constant coefficient two)**.** *Let $f = 2 + bx + P$ be nonconstant, where $b \ge 0$ and $P \in x^2\mathbb{N}_0[x]$. Put $m = P(1)$. Then $\mathsf{L}_+(f) \cap \{2, 3, \ldots\} = \{2, \ldots, K(f)\}$, where*

$$K(f) = \begin{cases} b + 1, & m = 0 \text{ or } 1, \\ b + 2, & m \ge 2. \end{cases}$$

The interval is empty when $K(f) = 1$.

Proof. Theorem 1.1 gives

$$K(f) = \min\left\{b + 2,\ b + 1 + \left\lfloor \frac{m}{2} \right\rfloor\right\}.$$ □

Example 4.2 (Why the linear coefficient matters)**.** The polynomials $2 + 2x$ and $2 + 2x^r$ for $r \ge 2$ have the same total coefficient sum, but

$$\mathsf{L}_+(2 + 2x) = \{2, 3\}, \qquad \mathsf{L}_+(2 + 2x^r) = \{2\}.$$

For the first, $2 + x + x$ is a three-atom representation. For the second, $U(f) = 2$ restricts every representation to at most two summands. Neither polynomial is an atom, since both have the nonunit constant factor 2.

Remark 4.3 (The constant-one boundary). The hypothesis $f(0) \ge 2$ cannot be omitted. For instance,

$$\mathsf{L}_+(1 + 7x + 6x^2) = \{2, 4, 5, 6, 7, 8\},$$

which has a gap. In any atomic representation of a constant-one polynomial, exactly one summand has positive constant coefficient; all others must be x. Thus a length-k representation of this example exists exactly when

$$1 + (8 - k)x + 6x^2$$

is an atom, with $1 \le k \le 8$. For a nonnegative integer t, the polynomial $1 + tx + 6x^2$ is reducible in $\mathbb{N}_0[x]$ precisely when

$$1 + tx + 6x^2 = (1 + ux)(1 + vx), \qquad uv = 6,$$

so $t = u + v \in \{5, 7\}$. These exclude exactly $k = 3, 1$, proving the claimed length set. The constant-one case therefore has a different character from Theorem 1.1.

5. A prescribed-length consequence of Laurent Goldbach theorems

Let

$$R_t = \mathbb{N}_0[x_1^{\pm 1}, \dots, x_t^{\pm 1}], \qquad t \geq 1.$$

Its units are precisely the coefficient-one monomials. A Laurent polynomial is a finite sum with exponent vectors in $\mathbb{Z}^t$; write $\nu(F) = F(1, \dots, 1)$ for its coefficient mass. Define $\mathsf{L}_{\pm}(F)$ to be the set of positive integers k for which F is a sum of exactly k multiplicative atoms of R_t.

Liao–Polo's univariate theorem [5], and its multivariate extension by Kaplan and Polo [3, Corollary 3.8], state that every $F \in R_t$ with at least two support exponents and $\nu(F) > 3$ is a sum of two atoms. The following is an elementary consequence, not a separate two-atom theorem.

Proposition 5.1 (Laurent prescribed lengths)**.** *Let $F \in R_t$ have at least two distinct support exponents. For every integer $k \geq 2$,*

$$k \in \mathsf{L}_{\pm}(F) \quad \Longleftrightarrow \quad 2k \leq \nu(F).$$

Length one occurs precisely when F itself is a Laurent atom.

Proof. Every Laurent atom has mass at least two, since every mass-one element is a unit. This proves necessity.

Suppose $\nu(F) \geq 2k$. Reserve one coefficient unit at each of two distinct support exponents. From the remaining units, select $2(k-2)$ and partition them into $k-2$ pairs. Each pair is an atom: it has mass two, whereas a product of two nonzero nonunits has mass at least four.

After removing these atoms, the remainder G has

$$\nu(G) = \nu(F) - 2(k-2) \geq 4,$$

and retains two support exponents because the reserved units were not removed. Kaplan–Polo's theorem expresses G as a sum of two Laurent atoms. Adjoining the removed $k-2$ atoms gives exactly k summands. □

Remark 5.2 (Why localization changes the answer). For nonmonomial Laurent polynomials, the admissible representation lengths are characterized entirely by the coefficient mass through the inequality $2k \leq \nu(F)$. In contrast, the ordinary polynomial semiring $\mathbb{N}_0[x]$ exhibits a richer structure. Every atom other than x contributes at least one unit to the constant coefficient, while each occurrence of x contributes to the linear coefficient. This introduces the additional sharp constraint $k \leq a + b$. Furthermore, since x is an atom of mass one in $\mathbb{N}_0[x]$ but becomes a unit in the Laurent semiring, the corresponding weighted constraint in the ordinary setting is $2k \leq f(1) + b$. Together, Theorem 1.1 and Proposition 5.1 provide a complete comparison of prescribed representation lengths in these two settings, identifying precisely how localization changes the additive structure of multiplicative atoms.

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